Ruiyang Jin

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Research Interests

My research lies at the intersection of *optimization* and *artificial intelligence (AI)* for complex, data-driven systems. I aim to develop efficient, theoretically grounded algorithms by integrating tools from optimization, data science, and AI, with a particular emphasis on *reinforcement learning, zeroth-order optimization, deep learning, and simulation-based optimization*. My work is motivated by real-world operational challenges, with primary applications in demand-side management for low-carbon energy systems and supply chain operations.

Education

09/2019–06/2024
Department of Industrial Engineering and Management, Peking University
Ph.D. in Management Science and Engineering Advisor: Prof. Jie Song

12/2022–12/2023
Department of Computing and Mathematical Sciences, California Institute of Technology
Visiting Scholar Advisor: Prof. Adam Wierman

09/2015–06/2019
College of Engineering, Peking University
B.E. in Aerospace Engineering

Appointments

10/2025–present
Hong Kong Space Robotics and Energy Center, Hong Kong University of Science and Technology
Associate Engineer

09/2024–present
Department of Systems Engineering, City University of Hong Kong
Postdoctoral Fellow Advisor: Prof. Siyang Gao

07/2024–09/2024
CityU Shenzhen Research Institute
Assistant Researcher

Publications (* denotes corresponding author)

Preprints and Ongoing Work

1. **Ruiyang Jin**, Yuke Zhou, Yujie Tang, Jie Song, Siyang Gao. Query-Efficient Zeroth-Order Algorithms for Nonconvex Optimization. *INFORMS Journal on Computing (major revision).*

2. Yuke Zhou, **Ruiyang Jin***, Siyang Gao, Jianxiao Wang, Jie Song. A Zeroth-Order Extra-Gradient Method for Black-Box Constrained Optimization. *Automatica (major revision).*
3. Ruipeng Jin, Yuke Zhou, Fan Zhang, **Ruiyang Jin***. From Data to Decisions: ICNN-Based Convex Optimization for Distributed Energy Management. *Computers and Operations Research (major revision).*
4. **Ruiyang Jin**, Siyang Gao, Henry Lam. A Min-Max Gradient Search Method for Constrained Simulation Optimization. *INFORMS Journal on Computing (under review).*
5. **Ruiyang Jin**, Jianxiao Wang, Fan Zhang, et al. Derivative-Free Optimization: Toward the Integration of Artificial Intelligence and Operations Research. *Nexus (under review).*
6. Zhengrun Wu, **Ruiyang Jin**, Jie Song, Guannan He. Zeroth-order Optimization Method for Decision-making of Energy Arbitrage with Degradation. *Fundamental Research (under review).*
7. **Ruiyang Jin**. Behavioral Common Random Numbers for LLM-Embedded Simulation Optimization, (*writing in progress*).

Journal Publications

1. **Ruiyang Jin**, Yujie Tang, Jie Song. Zeroth-Order Feedback-Based Optimization for Distributed Energy Management. *European Journal of Operational Research* (JCR Q1; ABS 4), 2026.
2. Rongjinzi Wang, **Ruiyang Jin***, Jie Song. Localized Multi-Agent Reinforcement Learning for Cooperative Management of Supply Chains. *Asia-Pacific Journal of Operational Research*, 2025, 2540010.
3. **Ruiyang Jin**, Yuke Zhou, Chao Lu, Jie Song. Deep Reinforcement Learning-based Strategy for Charging Station Participating in Demand Response. *Applied Energy* (JCR Q1; Nature Index), 2022, 328, 120140.
4. Pengya Wang, Jianxiao Wang, **Ruiyang Jin**, Gengyin Li, Ming Zhou. Integrating Biogas in Regional Energy Systems to Achieve Near-zero Carbon Emissions. *Applied Energy* (JCR Q1; Nature Index), 2022, 322, 119515.
5. Tiance Zhang, Jianxiao Wang, Hao Wang, **Ruiyang Jin**, Gengyin Li, Ming Zhou. On the Coordination of Transmission-Distribution Grids: A Dynamic Feasible Region Method. *IEEE Transactions on Power Systems* (JCR Q1), 2022, 38(2), 1857–1868.
6. **Ruiyang Jin**, Chao Lu, Jie Song. Manage Distributed Energy Storage Charging and Discharging Strategy: Models and Algorithms. *IEEE Transactions on Engineering Management* (JCR Q1), 2020, 69(3), 755–764.
7. **Ruiyang Jin**, Jie Song, Jie Liu, et al. Location and Capacity Optimization of Distributed Energy Storage System in Peak-Shaving. *Energies*, 2020, 13(3), 513.

Conference Proceedings

1. Yuke Zhou, **Ruiyang Jin***, Jianxiao Wang, Jie Song. Zeroth-Order Extra-Gradient Method for Constrained Convex Optimization. *The 64th IEEE Conference on Decision and Control (CDC 2025)*. IEEE, pp. 567–572.

2. **Ruiyang Jin**, Zaiwei Chen, Yiheng Lin, Jie Song, Adam Wierman. Approximate Global Convergence of Independent Natural Actor-Critic in Multi-Agent Systems. *The 28th International Conference on Artificial Intelligence and Statistics (AISTATS 2025)* (Top conference in Statistics and AI). PMLR 258, pp. 2818–2826.
3. Yuke Zhou, **Ruiyang Jin**, Jie Song. An online learning method for industrial demand response based on load disaggregation. *IEEE I&CPS Asia 2022*.
4. **Ruiyang Jin**, Yunlei Lu, Yunhong Wang, et al. The Short-Term Power Consumption Forecasting Based on the Portrait of Substation Areas. *2020 IEEE International Conference on Knowledge Graph (ICKG)*. IEEE, pp. 649–653.

Talks

1. Query-Efficient Zeroth-Order Algorithms for Black-Box Constrained Problems, *Academy of Mathematics and Systems Science*, 2026
2. Zeroth-Order Feedback-Based Optimization for Distributed Energy Management, *Annual Conference on Management Science and Engineering*, 2025
3. Randomized Zeroth-Order Feedback-Based Coordinate Descent Algorithm for Simulation-Embedded Problems, *INFORMS Annual Meeting*, 2024
4. Approximate Global Convergence of Independent Learning in Multi-Agent Systems, *SCIS Salon on Frontiers of Information Science and Technology*, 2024
5. Decentralized Zeroth-order Algorithms for Distributed Energy Systems, *Group Intelligence Seminar*, 2023
6. Reinforcement Learning and Its Application for Demand Response, *PKU*, 2021
7. Deep Reinforcement Learning-based Strategy for Charging Station Participating in Demand Response. *IEEE PES General Meeting*, 2021

Fundings

- "Algorithmic and Theoretical Research on Constrained Zeroth-Order Optimization in Complex Systems" (**PI**), Young Scientists Fund (Type C), *National Natural Science Foundation of China*, 300,000 RMB, 2026–2028
- "Modeling and Optimization of Distributed Energy Systems Considering Carbon Emissions" (**PI**), *Peking University*, 80,000 RMB, 2022–2024
- "Quantitative Research on the Economic and Environmental Value of Deep Peak-Shaving for Smart Power Plants in Zero-Carbon Energy Systems" (**PI**), *China Association for Science and Technology*, 50,000 RMB, 2022–2023

Honors and Awards

- **Selected for the Doctoral Dissertation Support Program:** Society of Management Science and Engineering of China, 2025
- **Outstanding graduates of Beijing (and Peking University):** Beijing Municipal Education Commission, 2024
- **BHP "Carbon and Climate" Weiming Scholar:** Peking University, 2022, among 10 selected graduate students of PKU

- **Best Student Paper Award:** IEEE I&CPS Asia, 2022
- **Young Talent in China High-Tech Innovation Think Tank,** China Association for Science and Technology, 2021
- **The 3rd China Industrial Engineering and Lean Management Innovation Competition:** Top prize, 2020, group leader
- **Peking University President Scholarship:** Peking University, 2020
- **Dean's scholarship of COE:** Peking University, 2020, 2023
- **Wu Kangming Scholarship**: Peking University, 2019
- **Yang Fuqing-Wang Yangyuan Scholarship**: Peking University, 2018
- **Wei Lin Scholarship**: Peking University, 2017

Membership and Service

- Reviewer for **Applied Energy, IEEE Transactions on Automation Science and Engineering, Operations Research, IEEE Transactions on Automatic Control,** etc.
- **Teaching Assistant** at PKU: Simulation and Modelling in the 2021 Fall
- **Teaching Assistant** at PKU: Applied Statistics in the 2020 Spring
- **Research mentor** for two undergraduate students in 2021 and 2022
- **Membership:** IEEE; INFORMS
- **Chairman of the Chinese calligraphy and painting association of PKU** in 2019